MATLAB Simulation Codes for [1, Sect. 4.2, 5.2, and 5.3]

These MATLAB codes reproduce the simulation results from [1, Sect. 4.2, 5.2, and 5.3]. In addition to the scripts `Ex1.m`, `Ex2.m`, and `Ex3.m`, the codes contain auxiliary functions `GammaSolver.m`, `L2polyfit.m`, `L2polyfit2.m`, `L2polyfitg.m`, `smac.m`, `smam.m`, and `waveform.m` for computing output injection gains and constructing optimal continuum approximations, spectral modal approximations, and aortic flow/pressure waveforms.The simulation results from [1, Sect. 4.2, 5.2, and 5.3] can be reproduced by running the script files which employ the auxiliary function files.

Description

The considered class of systems stems from a linearized model of blood flow in an artery arterial network of m arteries for $i \in \{1, \dots, m\}$ is of the form

$$\dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t), \tag{1a}$$
$$u_t^i(t,x) + \lambda^i(x) u_x^i(t,x) = w^i(x) v^i(t,x), \tag{1b}$$
$$v_t^i(t,x) - \mu^i(x) v_x^i(t,x) = \theta^i(x) u^i(t,x), \tag{1c}$$

with boundary conditions

$$u^i(t,0) = q^i v^i(t,0), \tag{2a}$$
$$v^i(t,1) = r^i u^i(t,1) + f^i \mathbf{C}\mathbf{X}(t), \tag{2b}$$

where (u^i, v^i) is the PDE state (in Riemann coordinates) and $\mathbf{C}\mathbf{X}(t)$ models aortic flow/pressure waveform. The measured output is the averaged peripheral flow/pressure which can be written as

$$Y(t) = \frac{1}{m} \sum_{i=m_1}^{m_2} g^i v^i(t,0), \tag{3}$$

for some weights g^i and integers $1 \le m_1 \le m_2 \le m$.

The developed observer design is based on the respective class of ODE - Continuum-PDE cascade systems is of the form

$$\dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t), \tag{4a}$$
$$u_t(t,x,y) + \lambda(x,y) u_x(t,x,y) = W(x,y) v(t,x,y), \tag{4b}$$
$$v_t(t,x,y) - \mu(x,y) v_x(t,x,y) = \theta(x,y) u(t,x,y), \tag{4c}$$

with boundary conditions

$$u(t,0,y) = Q(y) v(t,0,y), \tag{5a}$$
$$v(t,1,y) = R(y) u(t,1,y) + F(y) \mathbf{C}\mathbf{X}(t), \tag{5b}$$

and a weighted averaged measurement

$$Y(t) = \int_{y_1}^{y_2} g(y) v(t,0,y) dy, \tag{6}$$

over some $0 \leq y_1 < y_2 \leq 1$ with weight g. The respective (continuum) observer is of the form

$$\dot{\hat{\mathbf{X}}}(t) = \mathbf{A}\hat{\mathbf{X}}(t) + \mathbf{L}\left(\int_{y_1}^{y_2} g(y)\hat{v}(t,0,y)dy - Y(t)\right), \tag{7a}$$

$$\hat{u}_t(t,x,y) + \lambda(x,y)\hat{u}_x(t,x,y) = W(x,y)\hat{v}(t,x,y) + P_1(x,y)\left(\int_{y_1}^{y_2} g(y)\hat{v}(t,0,y)dy - Y(t)\right), \tag{7b}$$

$$\hat{v}_t(t,x,y) - \mu(x,y)\hat{v}_x(t,x,y) = \theta(x,y)\hat{u}(t,x,y) + P_2(x,y)\left(\int_{y_1}^{y_2} g(y)\hat{v}(t,0,y)dy - Y(t)\right), \tag{7c}$$

with boundary conditions

$$\hat{u}(t,0,y) = Q(y)\hat{v}(t,0,y), \tag{8a}$$

$$\hat{v}(t,1,y) = R(y)\hat{u}(t,1,y) + F(y)\mathbf{C}\hat{\mathbf{X}}(t). \tag{8b}$$

The gain $\mathbf{L} \in \mathbb{R}^n$ is chosen such that the matrix $\mathbf{A} + \mathbf{L}\int_{y_1}^{y_2} g(y)\boldsymbol{\gamma}_0(y)dy$ is Hurwitz, and $P_1, P_2 \in C([0,1]^2;\mathbb{R})$ are chosen as

$$P_1(x,y) = \boldsymbol{\gamma}^1(x,y)\mathbf{L}, \tag{9a}$$

$$P_2(x,y) = \boldsymbol{\gamma}^2(x,y)\mathbf{L}, \tag{9b}$$

where $\boldsymbol{\gamma}_0 \in C([0,1];\mathbb{R}^{1\times n})$ is the solution of the (Sylvester) equation

$$\lambda(x,y)\boldsymbol{\gamma}^1_x(x,y) + \boldsymbol{\gamma}^1(x,y)\mathbf{A} - W(x,y)\boldsymbol{\gamma}^2(x,y) = 0, \tag{10a}$$

$$\mu(x,y)\boldsymbol{\gamma}^2_x(x,y) - \boldsymbol{\gamma}^2(x,y)\mathbf{A} + \theta(x,y)\boldsymbol{\gamma}^1(x,y) = 0, \tag{10b}$$

with boundary conditions

$$Q(y)\boldsymbol{\gamma}^2(0,y) = \boldsymbol{\gamma}^1(0,y), \tag{11a}$$

$$R(y)\boldsymbol{\gamma}^1(1,y) + F(y)\mathbf{C} = \boldsymbol{\gamma}^2(1,y). \tag{11b}$$

The functions

```
[fpol, rd] = L2polyfit(dm,m,My)
[fpol, rd] = L2polyfitg(dm,m,My,m1,m2)
[fpol, rd] = L2polyfit2(fm,m,M,My,dx)
```

construct parameters for the continuum system (4), (5) such that they approximate the respective parameters of original system (1), (2) in the L^2 sense over the spatial domain. The functions `L2polyfit` and `L2polyfitg` construct 1-D polynomial of the form

$$f_{pol}(y) = \sum_{k=0}^{M_y} c_k L_k(y), \tag{12}$$

where L_k denotes Legendre polynomial of order k and the coefficients c_k are computed to provide the optimal L^2 fit to the original parameter data `dm`. The second output `rd` returns the L^2 residual of the fit. The function `L2polyfitg` is used for constructing the weight g appearing in (6), which additionally uses information about the summation bounds m_1, m_2 from (3) and the respective integration bounds y_1, y_2 (computed based on m_1, m_2) from (6). The function `L2polyfit2` is analogous to `L2polyfit` but acts in 2-D by constructing a polynomial of the form

$$f_{pol}(x,y) = \sum_{j=0}^{M_y} \sum_{i=0}^{M-j} c_{i,j} L_i(x) L_j(y), \tag{13}$$

where the coefficients $c_{i,j}$ are computed to provide the optimal L^2 fit to the original function data `fm`. The additional parameter `dx` enables computing the fit in the H^1 sense (in x), which is used for constructing λ and μ; see [1, Sect. 3.2] for details.

The function

```
[A, C, x0] = waveform(type)
```

constructs the ODE (1a) with initial condition $\mathbf{X}_0$ such that the output $\mathbf{C}\mathbf{X}(t)$ approximates the arterial flow (`type=Q`) or pressure (`type=P`) waveform according to [3, Table 6.1], or [3, Figure 6.1], respectively; see [1, (44)–(46)] for details.

The function

```
[G1s, G2s, rd] = GammaSolver(lam,mu,W,TH,Q,R,F,A,C,N,Ny)
```

computes an approximate solution $(\boldsymbol{\gamma}^1, \boldsymbol{\gamma}^2)$ to the Sylvester equation (10), (11) as power series

$$\boldsymbol{\gamma}^1(x,y) = \sum_{i=0}^{N_y} \sum_{j=0}^{N-i} c^1_{i,j} x^j y^i, \tag{14a}$$

$$\boldsymbol{\gamma}^2(x,y) = \sum_{i=0}^{N_y} \sum_{j=0}^{N-i} c^2_{i,j} x^j y^i, \tag{14b}$$

for given parameters (see, e.g., [2, 4] for background on the power series method). The third output `rd` returns the residual of the power series approximation.

The functions

```
[MM, KK, BB, Cv] = smam(m,Nx,lam,mu,W,TH,Q,R,F,g,m1,m2)
[MM, KK, BB, Cv, PP] = smac(Nx,Ny,lam,mu,W,TH,Q,R,F,g,g1,g2,L,G1,G2)
```

construct spectral modal approximations for the PDE parts of the system (1), (2) and the (continuum) observer (7), (8), respectively, for given parameters. The ODE resulting from the spectral approximation is of the form

$$\mathbf{M}\dot{\mathbf{c}}(t) = \mathbf{K}\mathbf{c}(t) + \mathbf{B}U(t), \tag{15a}$$

$$Y(t) = \mathbf{C}^v \mathbf{c}(t), \tag{15b}$$

where the vector $\mathbf{c} = (\mathbf{c}^u, \mathbf{c}^v)$ contains the coefficients for the spectral approximation of the PDE solution which is of the form

$$u(t,x,y) \approx \sum_{j=0}^{N_y} \sum_{i=0}^{N_x} c_{i,j}^u(t) L_i(x) L_j(y), \tag{16a}$$

$$v(t,x,y) \approx \sum_{j=0}^{N_y} \sum_{i=0}^{N_x} c_{i,j}^v(t) L_i(x) L_j(y), \tag{16b}$$

where the y-component is omitted in case of (1), (2). In case of the continuum observer, the additional output argument `PP` is a spectral approximation of the output injection operators P_1, P_2 given in (9). For more details on the spectral approximation, see [1, Sect. 4.1].

Examples

The simulation results for [1, Sect. 4.2], [1, Sect. 5.2], and [1, Sect. 5.3] can be reproduced by running the script `Ex1.m`, `Ex2.m`, and `Ex3.m`, respectively.

References

[1] J.-P. Humaloja and N. Bekiaris-Liberis. Observer design for a class of ODE – continuum-PDE cascade systems inspired by a control-theoretic model of large-scale arterial networks of blood flow. arXiv:2605.10524v2, 2026.

[2] J.-P. Humaloja and N. Bekiaris-Liberis. On computation of approximate solutions to large-scale backstepping kernel equations via continuum approximation. *Syst. Control Lett.*, 196:105982, 2025.

[3] N. Stergiopulos. *Computer simulation of arterial flow*. PhD thesis, Iowa State Univ., 1990.

[4] R. Vazquez, G. Chen, J. Qiao, and M. Krstic. The power series method to compute backstepping kernel gains: theory and practice. In *IEEE Conf. Decis. Control*, pages 8162–8169, 2023.